

HENRY GROVES

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Education

Further Study, University of Newcastle upon Tyne

2023-2027

- M.Sc.+B.Sc. (Hons) Computer Science

A Levels, Pontefract New College, Pontefract

2021-2023

- **A Levels:** Mathematics (**A***), Further Mathematics (**A**), Computer Science (**A**), Physics (**A**)
- **EPQ:** “To what extent will quantum computing replace high performance computing in computational biology?” (**A**)

GCSEs, Carleton High School, Pontefract, West Yorkshire

2016-2021

- GCSEs: 5 9's, 3 8's
- BTECs: D*2, D2

Work Experience

Northern Powergrid

Data Scientist (DSO)

May-September 2025

- Built a graph-based **simulation tool** to model high-voltage power networks, in both normal and abnormal conditions.
- Used historical and real-time data to simulate network and asset performance with a bottom-up approach, rather than a top-down approach previously used for such analysis.
- Tool uses future scenarios to account for the different ways the power grid may evolve looking to 2050.

Data Scientist (DSO)

May-September 2024

- Designed and architected an end-to-end **machine learning** solution to predict short-term distribution substation load.
- Built a **robust, efficient** and **scalable** data pipeline to ingest **half-hourly** meter readings from ~ 800 substations, filling gaps using AKIMA and lag-based methods.
- Used DTW K-Means **clustering** to find load profiles; proved that models trained on one member of the cluster can effectively predict all members.
- Designed and tuned both **TCN** and **BiTCN** neural-networks, for each cluster, with carefully selected **exogenous features** boosting model performance.
- Wrote a formal research paper outlining my work and findings to present at *CIRE*.

Data Analyst (System Forecasting)

May-August 2023

- Combined numerous data sources with **SQL** and **Python** scripting to design and implement a statistical model for predicting the demand of almost **4 million** customers on low-voltage electrical distribution networks.
- Created a dashboard in Excel from the output, and created visualizations for presentations to the wider team and executives.
- Extracted and processed data for, and collaborated with other colleagues on regulatory submissions under very tight deadlines whilst also working on the model.

Projects

“Biggmarket” - Item swapping app promoting sustainability

Built a full-stack item-swapping web application designed to promote sustainable consumption. Built using Next.js, TypeScript, tRPC, Tailwind CSS and backed by a PostgreSQL with Prisma, the app uses a “swipe left-right” interface, with real-time user messaging, authentication via Google OAuth, and a comprehensive user review system.

Conference/Oral Presentations

Robertson, B-A*; Jackson, L.; Groves, H.; and Hance, J.R. *Emotional Modulation of Temporal Order Memory in Childhood: A Quantum-Informed Account*. At Quantum Information and Probability: from Foundations to Engineering (QIP26), xx June 2026. Växjö, Sweden. *presenter

Key Skills

- Problem-Solving
- Attention to Detail
- Procedural Thinking
- Software Development
- Machine Learning and AI
- Data Exploration & Analysis

Programming Languages / Frameworks:

- Python 3
- Rust
- R
- numpy / pandas / scikit
- tensorflow / pytorch
- matplotlib / seaborn
- C/C++
- TypeScript
- React.JS / Next.JS
- Java

Additional Technologies:

- GNU/Linux
- LaTeX
- Git / GitHub
- SQL (PostgreSQL)
- MongoDB
- HTML/CSS
- tailwindcss
- Access / Excel

Languages:

- English (Native)
- German (B1)